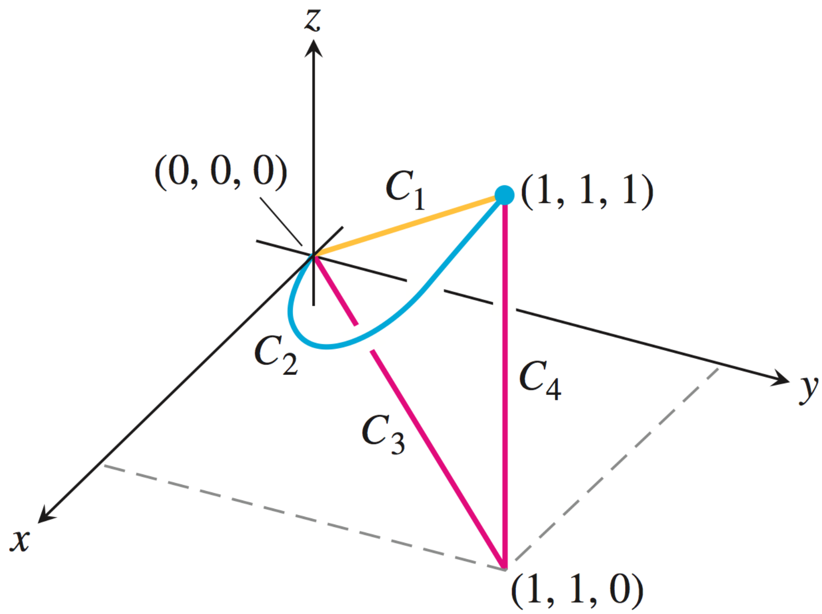
In addition to the online homework, please hand in the following problems:

Problem 1: Determine the work done in moving a mass through $\mathbf{F} = 3y\mathbf{i} + 2x\mathbf{j} + 4z\mathbf{k}$ from $(0,0,0)$ to $(1,1,1)$ along each of the following paths:

i. The straight-line C1

ii. The curved path C2 (intersection of 2 parabolic cylinders: $y = x^2$ and $z = y^2$)

iii. The path $C3 \cup C4$ consisting of the line segment from $(0,0,0)$ to $(1,1,0)$ followed by the segment from $(1,1,0)$ to $(1,1,1)$



Problem 2: Find the mass of a wire that lies along the curve $\mathbf{r}(t) = (t^2 - 1)\mathbf{j} + 2t\mathbf{k}$, $0 \le t \le 1$ if the density is given by $\delta(x,y,z) = \frac{5}{2}z$

Problem 3: Is there a direction $\mathbf{u}$ in which the rate of change of the temperature function $T(x,y,z) = 2xy - yz$ at $P(1,-1,0)$ equals $-3°C/ft$? If so, find it. If not, say why.

Problem 4: Prove the following properties of gradients:

a. $\nabla(kf) = k\nabla f$ for k constant

b. $\nabla\left(\frac{f}{g}\right) = \frac{\nabla f \cdot g - f \cdot \nabla g}{g^2}$